DS&AI Lab Project [Term Jan 2026]

Action-Oriented Indoor Navigation Assistance for the Visually Impaired

MILESTONE-2: Dataset Identification & Preparation

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GROUP: 4

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1. Introduction & Objectives

The objective of Milestone-2 is to identify, evaluate, and prepare datasets required for training the perception and interaction modules of the navigation assistance system.
The datasets used in this project support three primary components of the system:

1. **Object Detection Module** – detects indoor obstacles and objects using RGB images.
2. **Depth Estimation Module** – estimates distance to objects from monocular camera images.
3. **Speech / TTS Module** – converts navigation decisions into spoken instructions.

The goal of this milestone is to ensure that the data used for these modules is:

- Clearly sourced and legally usable
- Properly structured and formatted
- Sufficient in quantity and diversity
- Free from significant quality issues
- Prepared in a reproducible preprocessing pipeline
- Split correctly to avoid data leakage

2. Data-Specific Scope & Assumptions

In-Scope	Out-of-Scope
Indoor environment datasets	Outdoor navigation datasets
Near-field obstacle detection (0–3 meters)	Large-scale SLAM map datasets
RGB camera-based perception	Long-range depth estimation (>5 meters)
Depth estimation from monocular or RGB-D datasets	GPS-based navigation
Speech synthesis datasets for navigation instructions	

Table 2.1: Data-Specific Scope & Assumptions

3. Data Sources, Ownership & Usage Constraints

Table 3.1 summarizes the datasets identified for each module. These datasets were selected based on their relevance to indoor navigation tasks, availability for research use, and compatibility with our model architectures.

The datasets include a mix of public benchmark datasets, open-source datasets, and custom collected data to ensure both diversity and relevance to the final deployment scenario.

Module	Dataset	Type	Approx Size	Purpose
Depth	NYU Depth V2	RGB-D Indoor	~1.4k labelled pairs	Indoor depth benchmark
Depth	ScanNet	RGB-D video dataset	~2.5M frames	Large-scale indoor scenes
Depth	DIODE	RGB + LiDAR depth	~12k images	Depth generalization
Depth	Matterport3D	RGB-D panoramic	~90 buildings	Indoor geometry diversity
Depth	SUN RGB-D	RGB-D dataset	~10k images	Indoor scene depth
Detection	MS COCO	Object detection	~118k images	Base training dataset
Detection	Open Images	Object detection	~9M images	Large-scale object diversity
Detection	Indoor Object Dataset (Kaggle)	Indoor images	~20k images	Indoor obstacle detection
Detection	Ultralytics Dataset	YOLO formatted dataset	~50k images	Model fine-tuning
Detection	Custom Indoor Dataset	RGB images	Target 1k–2k images	Deployment environment
Speech	LJ Speech	Speech synthesis	13k clips	TTS baseline
Speech	LibriTTS	Speech corpus	585 hours	Voice modeling
Speech	Common Voice	Multilingual speech	20k+ hours	Language diversity
Speech	VCTK	Multi-speaker speech	44 hours	Speaker variation
Speech	Custom Navigation Commands	Short phrases	~50 commands	System evaluation

Table 3.1: Dataset Overview

These datasets together provide a balance of **scale, diversity, and task relevance** necessary for training and evaluating the navigation system.

4. Data Sources, Ownership & Usage Constraints

Each dataset used in this project is sourced from publicly available repositories or created internally.

Depth Datasets

1. **NYU Depth V2**
    a. Source: NYU Vision Lab
    b. License: Research use
    c. Format: RGB images + aligned depth maps
2. **ScanNet**
    a. Source: Princeton Vision & Learning Lab
    b. License: Academic research license
    c. Format: RGB-D video frames
3. **DIODE**
    a. Source: University of Maryland
    b. License: Open research dataset
    c. Format: RGB images with LiDAR-based depth
4. **Matterport3D**
    a. Source: Matterport Research
    b. License: Academic use
    c. Format: RGB-D panoramic images
5. **SUN RGB-D**
    a. Source: MIT / Princeton
    b. License: Research use
    c. Format: RGB + depth sensor data

Object Detection Datasets

1. **MS COCO**
    a. Source: Microsoft
    b. License: Creative Commons
    c. Format: COCO JSON annotations
2. **Open Images Dataset**
    a. Source: Google
    b. License: Open dataset
    c. Format: Bounding boxes + labels
3. **Indoor Object Dataset (Kaggle)**
    a. Source: Kaggle community
    b. Format: Annotated indoor scenes
4. **Ultralytics YOLO Dataset**
    a. Source: Ultralytics training datasets
    b. Format: YOLO annotation format

5. **Custom Indoor Dataset**
    a. Collected using our prototype camera setup
    b. Target scenes include hallways, rooms, furniture layouts

Speech / TTS Datasets

1. **LJ Speech**
    a. Single speaker English dataset used for TTS training.
2. **LibriTTS**
    a. Derived from LibriSpeech with high-quality recordings.
3. **Common Voice**
    a. Mozilla multilingual speech dataset.
4. **VCTK**
    a. Multi-speaker English speech corpus.
5. **Custom Navigation Command Dataset**
    a. Recorded navigation instructions such as:
    b. "Obstacle ahead"
    c. "Turn left"
    d. "Move forward"
    e. "Step down"

5. Detailed Dataset Description

Depth Dataset Characteristics

Depth datasets contain RGB images paired with depth maps representing distance information for each pixel. These datasets capture indoor environments such as living rooms, kitchens, offices, and hallways.

Typical properties include:

- Resolution ranging from 480×640 to 1024×768
- Depth values measured in meters
- Sensor noise in reflective or transparent surfaces
- Missing depth values in some pixels

Object Detection Dataset Characteristics

Object detection datasets contain images annotated with bounding boxes around objects.

Target indoor object classes include

Chair	Couch	Bed	Table	Desk	Cabinet	Bookshelf
Bench	Door	Window	Stairs	Wall	Floor	Refrigerator
Microwave	Oven	Toaster	Sink	TV	Backpack	Suitcase
Handbag	Bottle	Cup	Plant	Vase	Person	Dog

Matrix 5.1: Dataset Classes ▪ Furniture ▪ Navigation Elements ▪ Appliances ▪ Indoor Objects ▪ Living Beings

6. Dataset Quality Assessment

Dataset quality is evaluated using the following criteria:

Noise

Some images may have induced noise in them, such as rain, blur, discolored pixels, etc

Duplicate Images

Video-based datasets may contain redundant frames. Frame filtering will remove near-identical frames to prevent training bias.

Class Imbalance

Some objects appear more frequently than others. Class distribution will be analyzed and balanced through sampling or augmentation.

7. Dataset Adequacy & Expansion Strategy

Although large datasets exist, some may not fully represent the real deployment environment.

To address this:

- Additional **custom indoor images** will be collected.
- Data augmentation techniques will be applied.

Data Augmentation Methods

- brightness adjustments
- contrast changes
- gaussian noise
- slight rotations
- scaling

Horizontal flipping will be used cautiously because it can alter spatial semantics such as left/right navigation.

8. Train / Validation / Test Split Strategy

Datasets will be split using the following proportions:

- **Training set:** 70%
- **Validation set:** 15%
- **Test set:** 15%

The training set will be used for model learning and parameter optimization. The validation set will be used during training to monitor model performance and tune hyperparameters, while the test set will be kept completely unseen and used only for final evaluation. To avoid data leakage, images from the same scene or sequence will be placed in the same split to ensure that similar frames do not appear across different datasets.

9. Image Cleaning and Preprocessing

Before training the perception models, the collected images will undergo a cleaning and preprocessing stage to improve overall data quality and robustness of the models. Since images captured in real-world environments may contain **noise, blur, rain**-like artifacts, or **lighting inconsistencies**, several preprocessing techniques will be applied.

Denoising

Images captured in low-light conditions or from inexpensive sensors may contain random noise. Image denoising techniques help remove these unwanted distortions while preserving important object features. Models such as ***DnCNN, Noise2Noise***, and ***FFDNet*** can be used for effective image denoising.

Deblurring

Motion blur may occur due to camera movement or user motion while capturing images. Deblurring methods help restore the sharpness of objects and improve feature visibility. Deep learning models such as ***DeblurGAN, SRN-DeblurNet***, and ***MPRNet*** are commonly used for image deblurring tasks.

Deraining / Artifact Removal

In some cases, images may contain rain-like streaks or visual artifacts caused by environmental conditions or sensor issues. Image deraining models such as ***RESCAN, DerainNet***, and ***PReNet*** can be applied to remove these artifacts and improve image clarity.

Additional Image Enhancement

Additional preprocessing techniques such as **contrast normalization, brightness adjustment, image resizing, and color normalization** will be applied to ensure consistent image quality across the dataset.

To implement these preprocessing steps, several deep learning–based image restoration models will be explored, including ***DnCNN, DeblurGAN, MPRNet*, and *RESCAN***. These models help improve the quality of input images before they are passed into the object detection and depth estimation pipelines, thereby improving the robustness and accuracy of the overall system.